Sr.No. 8507

Bharatiya Vidya Bhavan's

SARDAR PATEL INSTITUTE OF TECHONOLOGY

(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058

MID SEMESTER EXAMINATION

AY: 2025-26 | Session: Odd MSE October 2025 | Date: 08-10-2025
Sem: 1 | Slot: 10:00am - 11:00am | C. Code: CS413
Exam Type: Written | Block:401 | Course Section:1 | Seat No: 2022600037
L_29_494_986_1_1505_38370

Name of the candidate. Om Mukund Pandey

Candidate's UID number:

2	0	2	2	6	0	0	0	3	7

Examination class room number:

4	0	1

Invigilator's signature with date: 8/10/25

(To be filled in by the candidate)

EXAMINATION: BE BTECH
(For example FY MCA/BE BTECH)

SEMESTER : VII

I / II / III / IV / V / VI / VII / VIII

PROGRAM: CSE-(AI-ML)

DAY AND DATE: Wednesday, 08/10/25

COURSE NAME: Deep Learning

(To be filled in by the examiner)

Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.

2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.

3) Do not write anything in the column provided for the marks to be assigned by the examiners.

4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.

5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.

6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.

7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.

8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting.**

9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.

10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.

11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.

12) The answer-books will be scrutinized before they are sent to examiners.

13) All answer-books supplied shall be returned whether written or blank.

14) Nothing shall be written on the question-paper.

15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.

16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.

17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.

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Space For Marks	Question No.	
	Q.1. a.	

Ans:- $f(w) = \frac{1}{2} \|Xw - y\|^2$

Gradient Descent $= \eta \frac{\partial E}{\partial w}$

$= \frac{\eta \partial f(w)}{\partial w} = \eta \frac{\partial}{\partial(w)} \frac{1}{2} \|Xw - y\|^2$

$= \eta \frac{1}{\cancel{2}} \times \cancel{2} X \|Xw - y\|$

$= \eta X \|Xw - y\|$

Eigen Value of $X^T X$ would be

$X^T X - \lambda I = 0$

Let $x = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

$X^T X - \lambda = 0$

$\begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} - \lambda = 0$

$\begin{vmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix} = 0 \quad (2-\lambda)^2 - 4 = 0$

$\lambda^2 - 4\lambda = 0$

$\lambda = 0, 4 \quad \therefore \lambda_{max} = 4$

$\frac{2}{\lambda_{max}} = \underline{0.5}$

for converging gradient descent in $\eta X \| Xw - y \|$

$\eta \leq 0.5$ or $\frac{2}{\lambda_{max}}$.

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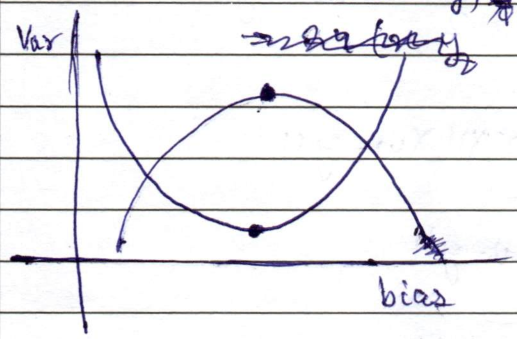
Q.1.
b.

Ans:- Bias Variance ~~(tradeoff)~~ decomposition for the expected square error predictor.

$$\frac{\partial}{\partial x \partial y}(x-y)^2 = \frac{\partial}{\partial x}(x-y)^2 \; \frac{\partial}{\partial y}(x-y)^2$$

$$= 2(x-y) \cdot -2(x-y)$$



when a model capacity is too high, overfitting occurs.

Overfitting occurs when the model also gets trained ~~(on)~~ on the noise of the data.

In such cases the model will give amazing results during testing but will fail during validation.

This affects the model during prediction as it too complex and gives output also based on noise & outliers thereby affecting generalisation.

~~To avoid avoid overfitting, a model must~~)

We can avoid overfitting by :-
1. Dropout
2. Feature selection
3. Data preprocessing
4. Reducing complexity
5.

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Space For Marks	Question No.	
	Q.2.	
	Q.	
	Ans:-	Computational complexity of

$H \times W \times C$ with K filters of $F \times F$

the 1 $F \times F$ filter would fit how many times of the $H \times W$ 2D image

$(H - F + 1) \times (W - F + 1)$ times

multiplied by K such filters of

$(H - F + 1) \times (W - F + 1) \times K$.

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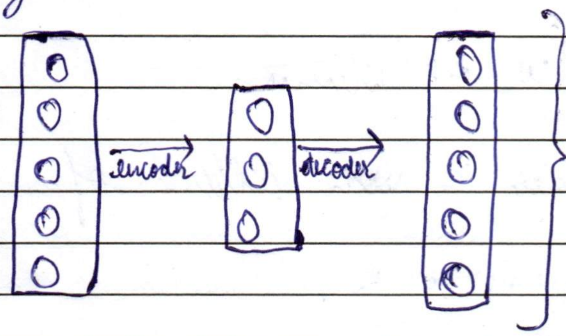
Q.2. b.

Ans:- In a linear undercomplete autoencoder, trained with MSE, the middle layer has a smaller size than the input layer.

Data / Dimensionality is reduced and then is again reconstructed in the output.



→ Undercomplete Encoder.

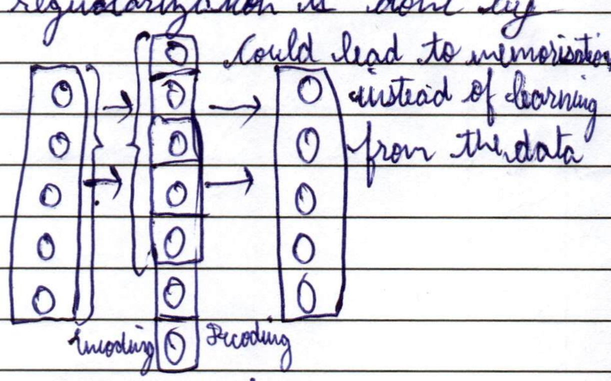
Here the encoder reduces the dimensionality of the input data and creates this subspace exactly equivalent to a PCA.

$$MSE = \frac{1}{n} \sum_{d=1}^{n} (t_d - o_d)^2$$

An overcomplete ~~en~~ encoder without regularization tends to (~~learn~~) learn the identity function as the size of the middle layer is larger than the size of the input layer. This may lead to the encoder directly copy pasting data.

To avoid such instances, regularization is done by methods like :-

~~1. Sparse Encoding~~
1. Sparsity AE
2. Variation AE
3. Noising / Denoising AE
4. Contractive AE.



Overcomplete Encoder

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	Q3.	
	~~A~~ a.	
	Ans-	~~A~~ As we find the recursive gradient in an RNN via chain rule, the gradient keeps on reducing and vanishes as it tends ~~to~~ to 0.
		The LSTM has a forget gate by which it solves this problem by allowing the model to remember and forget accordingly.

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Q.3.
b.

Ans:- GRU (Gated Recurrent) :-

GRU has 2 gates, Update Gate & Reset Gate.

$U_t = \sigma(W_u[x_t, h_{t-1}] + b_u)$
$r_t = \sigma(W_r[x_t, h_{t-1}] + b_r)$
$\tilde{h}_t = \tanh(W_h[x_t, r_t * h_{t-1}] + b_h) \rightarrow$ Candidate State
$h_t = \cancel{(1-h).r_t)}\ \tilde{h}_t . U_f + (1-h_{t-1}).r_f$

Hence we can clearly see how the update rule is a convex combination of previous hidden state & candidate state.

GRU have 3 parameters, i.e. update gate, reset gate & candidate state

LSTM has 4 parameters, i.e. input gate, forget gate, output gate & candidate state.

As GRU have lesser parameters, they are comparatively lighter than LSTM. This reduces their requirement of compute power.

In scenarios where the data size is huge, GRU would make a significant difference as lesser compute would be require due to less number of parameters and calculations

Real world examples are of big data analytics where complex patterns are required to be found while still remembering the input data.

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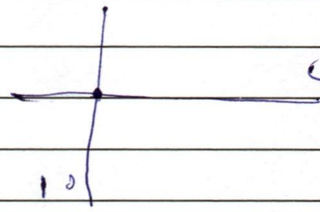
$\frac{1}{n} \sum (td^{2} - od^{2})^2$ $\quad (x-y)^2$

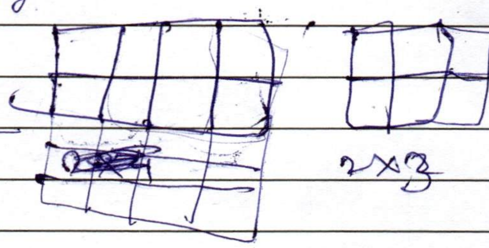
$\frac{\partial}{\partial x} 2(x-y) \quad \frac{\partial}{\partial y} -2(x-y)$

$\frac{1}{n} \Sigma (x-y)^2$

$\frac{1}{n} \Sigma 2(x-y)$

$\frac{2}{n} (x-y)$





2×3

1 0 1 0 1 0

0 1 0 1 0

2 2 1 0

2 2 0 1

$\begin{vmatrix} 2-\lambda & 2 \\ 2 & 2-\lambda \end{vmatrix}$

5×4

$(W-1) \times (H-1)$